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Tracking Device Location Identification

Abstract

A method includes receiving, by a tracking system server, a stream of tracking device event information associated with a plurality of tracking devices, extracting, by the tracking system server, and from the stream of tracking device event information, tracking device event information including location updates, filtering, by the tracking system server, the tracking device event information by removing information unrelated to any calculation of a last known location of at least one tracking device of the plurality of tracking devices, associating, by the tracking system server, the filtered tracking device event information with one or more disconnection events associated with the at least one tracking device, and identifying, by the tracking system server, and based at least in part on the filtered tracking device event information and the one or more disconnection events, the last known location of the at least one tracking device.

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Background/Summary

PRIORITY [0001] This application is a continuation under 35 U.S.C. § 120 of U.S. application Ser. No. 18/426,065, filed 29 Jan. 2024, which is a continuation under 35 U.S.C. § 120 of U.S. application Ser. No. 18/067,663, filed 16 Dec. 2022, now U.S. Pat. No. 11,889,375, which is a continuation under 35 U.S.C. § 120 of U.S. application Ser. No. 17/379,744, filed 19 Jul. 2021, now U.S. Pat. No. 11,533,581, which is a continuation under 35 U.S.C. § 120 of U.S. application Ser. No. 16/843,877, filed 8 Apr. 2020, now U.S. Pat. No. 11,070,934, which is a continuation under 35 U.S.C. § 120 of U.S. application Ser. No. 16/703,553, filed 4 Dec. 2019, now U.S. Pat. No. 10,645,526, which is a continuation under 35 U.S.C. § 120 of U.S. application Ser. No. 16/525,182, filed 29 Jul. 2019, now U.S. Pat. No. 10,536,798, which is a continuation under 35 U.S.C. § 120 of U.S. application Ser. No. 16/228,474, filed 20 Dec. 2018, now U.S. Pat. No. 10,412,540, which is a continuation under 35 U.S.C. § 120 of U.S. application Ser. No. 15/897,159, filed 15 Feb. 2018, now U.S. Pat. No. 10,219,107, which claims the benefit of U.S. Provisional Application No. 62/460,602, filed 17 Feb. 2017, each of which is incorporated herein by reference.

TECHNICAL FIELD
[0002] This disclosure relates generally to tracking devices, and more specifically, to determining a last known location for a tracking device.
BACKGROUND
[0003] Electronic tracking devices have created numerous ways for people to track the locations of people and/or objects. For example, a user can use GPS technology to track a device remotely or determine a location of the user. In another example, a user can attach a tracking device to an important object, such as keys or a wallet, and use the features of the tracking device to more quickly locate the object, (e.g., if it becomes lost).
[0004] However, traditional tracking devices and corresponding systems suffer from one or more disadvantages. For instance, if a tracking device is lost, the limited wireless range of the tracking device prevents an owner of the tracking device from locating the tracking device from outside the range of the tracking device. Generating an accurate last known location of a tracking device can

be important in allowing an owner to find a lost tracking device, for example, by narrowing the area to search for the tracking device. Existing methods for determining a last known location do not incorporate all available information to estimate the last known location, therefore improved methods for last known location determination are needed.

SUMMARY OF PARTICULAR EMBODIMENTS

[0005] A tracking system can coordinate multiple search party members in order to locate a lost tracking device. Upon losing a tracked item, the owner of the lost tracking device can request that the tracking system start a search party to locate the lost item. The tracking server can utilize information provided in this request to establish a set of search party criteria and coordinate a search for the lost item. In some implementations, search party criteria include characteristics of desired search party members, such as a current proximity to the last known location of the lost item. Using these criteria, the tracking system can then select and invite suitable candidates out of the community of users of the tracking system to participate in searching for the lost tracking device.

[0006] Candidates who accept the invitation to the search party are added to the search party and can be provided with the last known location of the lost tracking device. Based on this information, the search party members can begin looking for the lost tracking device. When one of the candidate search party members finds the lost tracking device or detects its signal, its location can then be provided back to the tracking system, which can in turn relay this information back to the owner of the lost tracking device. This allows the tracking system to leverage a community of users in locating a lost tracking device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. **1** illustrates an example tracking system environment in which a tracking device can operate, according to one embodiment.

[0008] FIG. **2** illustrates an example tracking system for use in a tracking system environment, according to one embodiment.

[0009] FIG. **3** illustrates an example user mobile device for use in a tracking system environment, according to one embodiment.

[0010] FIG. **4** illustrates an example community mobile device for use in a tracking system environment, according to one embodiment.

[0011] FIG. **5** illustrates an example tracking device for use in a tracking system environment, according to one embodiment.

[0012] FIG. **6** illustrates an example environment for determining a tracking device location based on later mobile device location updates, according to one embodiment.

[0013] FIG. **7** illustrates an example environment for determining a tracking device location based on multiple reported locations, according to one embodiment.

[0014] FIG. **8** is a flowchart illustrating an example process for filtering data for determining a last known location of a tracking device, according to one embodiment.

[0015] FIG. **9** is a flowchart illustrating an example process for determining the last known location of a tracking device, according to one embodiment.

[0016] The figures depict various embodiments of the present invention for purposes of illustration only. One skilled in the art will readily recognize from the following discussion that alternative embodiments of the structures and methods illustrated herein may be employed without departing from the principles of the invention described herein.

DESCRIPTION OF EXAMPLE EMBODIMENTS

Environment Overview

[0017] Embodiments described herein detail functionality associated with a tracking device. A user can attach a tracking device to or enclose the tracking device within an object, such as a wallet, keys, a car, a bike, a pet, or any other object that the user wants to track. Or, a tracking device can be a device with a primary purpose unrelated to tracking functionality (e.g., a set of headphones, an electronic key, a wireless speaker, a fitness tracker, a camera) that has an integrated tracking component that allows the device to be tracked. The user can then use a mobile device (e.g., by way of a software application installed on the mobile device) or other device or service to track the tracking device. For example, the mobile device can perform a local search for a tracking device. However, in situations where the user is unable to locate the tracking device using their own mobile device (e.g., if the tracking device is beyond a distance within which the mobile device and the tracking device can communicate), the user can leverage the capabilities of a community of users of a tracking device system as described below.

[0018] A tracking system (also referred to herein as a “cloud server,” “tracking server,” or simply “server”) can maintain user profiles associated with a plurality of users of the tracking system. The tracking system can associate each user within the system with one or more tracking devices associated with the user (e.g., tracking devices that the user has purchased and is using to track objects owned by the user, or devices that include a tracking component and have additional non-tracking features). If the user's tracking device, or the object to which the tracking device is attached, becomes lost or stolen, the user can send an indication that the tracking device is lost to the tracking system, which is in communication with one or more mobile devices associated with the community of users in communication with the system. The tracking system can set a flag indicating the tracking device is lost. When one of a community of mobile devices that are scanning for nearby tracking devices and providing updated locations to the tracking system identifies a flagged tracking device, the tracking system can associate the received location with the flagged tracking device, and relay the location to a user of the tracking device, thereby enabling the user to locate the lost tracking device. As used herein, “mobile device” can refer to a phone, tablet computer, or other connected device, and can also refer to systems typically not consider mobile, such as servers, routers, gateways, access points, and specialized systems configured to couple to tracking devices and report a location of the tracking devices.

[0019] As used herein, “tracking device” can refer to any device configured to communicate with another device for the purpose of locating the tracking device. Tracking devices can be specialized or single-purpose devices (e.g., self-contained devices that include circuitry or components to communicate with another device). However, “tracking device” as used herein can also refer to device or object with a different primary function but with secondary tracking device functionality. For example, a wireless speaker can include tracking device components that allow a user to track and/or locate the wireless speaker. In some embodiments, a tracking device platform can be established such that devices and objects that satisfy one or more criteria can act as tracking devices within a tracking device ecosystem. For instance, a tracking device provider can provide an SDK or custom chipset that, when incorporated into an object or device, enable the object or device to function as tracking devices, to communicate with other devices within the tracking device ecosystem, and to implement the functionalities described herein.

[0020] FIG. **1** illustrates an example tracking system environment in which a tracking device can operate, according to one embodiment. The environment of FIG. **1** includes a tracking system **100** communicatively coupled to a mobile device **102** associated with the user **103** via a first network **108**. The tracking system **100** is also communicatively coupled to a plurality of community mobile devices **104***a* through **104***n* (collectively referred to herein as “community mobile devices **104**”) associated with a plurality of users **105***a* through **105***n* of the tracking system **100** (collectively referred to herein as “community users **105**”) via the first network **108**. As will be explained in more detail below, the tracking system **100** can allow the user **103** to manage and/or locate a tracking device **106** associated with the user **103**. In some embodiments, the tracking system **100** leverages the capabilities of community mobile devices **104** to locate the tracking device **106** if the location of the tracking device is unknown to the user **103** and beyond the capabilities of mobile device **102** to track. In some configurations, the user **103** may own and register multiple tracking devices **106**. Although FIG. **1** illustrates a particular arrangement of the tracking system **100**, mobile device **102**, community mobile devices **104**, and tracking device **106**, various additional arrangements are possible.

[0021] In some configurations, the user **103** may be part of the community of users **105**. Further, one or more users **105** may own and register one or more tracking devices **106**. Thus, any one of the users within the community of users **105** can communicate with tracking system **100** and leverage the capabilities of the community of users **105** in addition to the user **103** to locate a tracking device **106** that has been lost.

[0022] The tracking system **100**, mobile device **102**, and plurality of community mobile devices **104** may communicate using any communication platforms and technologies suitable for transporting data and/or communication signals, including known communication technologies, devices, media, and protocols supportive of remote data communications.

[0023] In certain embodiments, the tracking system **100**, mobile device **102**, and community mobile devices **104** may communicate via a first network **108**, which may include one or more networks, including, but not limited to, wireless networks (e.g., wireless communication networks), mobile telephone networks (e.g., cellular telephone networks), closed communication networks, open communication networks, satellite networks, navigation networks, broadband networks, narrowband networks, the Internet, local area networks, and any other networks capable of carrying data and/or communications signals between the tracking system **100**, mobile device **102**, and community mobile devices **104**. The mobile device **102** and community of mobile devices **104** may also be in communication with a tracking device **106** via a second network **110**. The second network **110** may be a similar or different type of network as the first network **108**. In some embodiments, the second network **110** comprises a wireless network with a limited communication range, such as a Bluetooth or Bluetooth Low Energy (BLE) wireless network. In some configurations, the second network **110** is a point-to-point network including the tracking device **106** and one or more mobile devices that fall within a proximity of the tracking device **106**. In such embodiments, the mobile device **102** and community mobile devices **104** may only be able to communicate with the tracking device **106** if they are within a close proximity to the tracking device, though in other embodiments, the tracking device can use long-distance communication functionality (for instance, a GSM transceiver) to communicate with either a mobile device **102**/**104** or the tracking system **100** at any distance. In some configurations, the mobile device **102** and one or more community mobile devices **104** may each be associated with multiple tracking devices associated with various users.

[0024] As mentioned above, FIG. **1** illustrates the mobile device **102** associated with the user **103**. The mobile device **102** can be configured to perform one or more functions described herein with respect to locating tracking devices (e.g., tracking device **106**). For example, the mobile device **102** can receive input from the user **103** representative of information about the user **103** and information about a tracking device **106**. The mobile device **102** may then provide the received user information, tracking device information, and/or information about the mobile device **102** to the tracking system **100**. Accordingly, the tracking system **100** is able to associate the mobile device **102**, the user **103**, and/or the tracking device **106** with one another. In some embodiments, the mobile device **102** can communicate with the tracking device **106** and provide information regarding the location of the tracking device to the user **103**. For example, the mobile device **102** can detect a communication signal from the tracking device **106** (e.g., by way of second network **110**) as well as a strength of the communication signal or other measure of proximity to determine an approximate distance between the mobile device **102** and the tracking device **106**. The mobile device **102** can then provide this information to the user **103** (e.g., by way of one or more graphical user interfaces) to assist the user **103** to locate the tracking device **106**. Accordingly, the user **103** can use the mobile device **102** to track and locate the tracking device **106** and a corresponding object associated with the tracking device **106**. If the mobile device **102** is located beyond the immediate range of communication with the tracking device **106** (e.g., beyond the second network **110**), the mobile device **102** can be configured to send an indication that a tracking device **106** is lost to the tracking system **100**, requesting assistance in finding the tracking device. The mobile device **102** can send an indication of a lost device in response to a command from the user **103**. For example, once the user **103** has determined that the tracking device **106** is lost, the user can provide user input to the mobile device **102** (e.g., by way of a graphical user interface), requesting that the mobile device **102** send an indication that the tracking device **106** is lost to the tracking system **100**. In some examples, the lost indication can include information identifying the user **103** (e.g., name, username, authentication information), information associated with the mobile device **102** (e.g., a mobile phone number), information associated with the tracking device (e.g., a unique tracking device identifier), or a location of the user (e.g., a GPS location of the mobile device **102** at the time the request is sent).

[0025] The tracking system **100** can be configured to provide a number of features and services associated with the tracking and management of a plurality of tracking devices and/or users associated with the tracking devices. For example, the tracking system **100** can manage information and/or user profiles associated with user **103** and community users **105**. In particular, the tracking system **100** can manage information associated with the tracking device **106** and/or other tracking devices associated with the user **103** and/or the community users **105**.

[0026] As mentioned above, the tracking system **100** can receive an indication that the tracking device **106** is lost from the mobile device **102**. The tracking system **100** can then process the indication in order to help the user **103** find the tracking device **106**. For example, the tracking system **100** can leverage the capabilities of the community mobile devices **104** to help find the tracking device **106**. In particular, the tracking system **100** may set a flag for a tracking device **106** to indicate that the tracking device **106** lost and monitor communications received from the community mobile devices **104** indicating the location of one or more tracking devices **106** within proximity of the community mobile devices **104**. The tracking system **100** can determine whether a specific location is associated with the lost tracking device **106** and provide any location updates associated with the tracking device **106** to the mobile device **102**. In one example, the tracking system may receive constant updates of tracking device **106** locations regardless of whether a tracking device **106** is lost and provide a most recent updated location of the tracking device **106** in response to receiving an indication that the tracking device **106** is lost.

[0027] In some configurations, the tracking system **100** can send a location request associated with the tracking device **106** to each of the community mobile devices **104**. The location request can include any instructions and/or information necessary for the community mobile devices **106** to find the tracking device **102**. For example, the location request can include a unique identifier associated with the tracking device **106** that can be used by the community mobile devices **104** to identify the tracking device **106**. Accordingly, if one of the community mobile devices **104** detects a communication from the tracking device **106** (e.g., if the community mobile device **104** is within range or moves within range of the communication capabilities of the tracking device **106** and receives a signal from the tracking device **106** including or associated with the unique identifier associated with the tracking device **106**), the community mobile device **104** can inform the tracking system **100**. Using the information received from the community mobile devices **104**, the tracking system **100** can inform the user (e.g., by way of the mobile device **102**) of a potential location of the tracking device **106**.

[0028] As shown in FIG. **1** and as mentioned above, the tracking system **100** can communicate with a plurality of community mobile devices **104** associated with corresponding community users **105**. For example, an implementation may include a first community mobile device **104***a* associated with a first community user **105***a*, a second community mobile device **104***b* associated with a second community user **105***b*, and additional communication mobile devices associated with additional community users up to an nth community mobile device **104***n* associated with an nth community user **105***n*. The community mobile devices **104** may also include functionality that enables each community mobile device **104** to identify a tracking device **106** within a proximity of the community mobile device **104**. In one example, a first community mobile device **104***a* within proximity of a tracking device **106** can communicate with the tracking device **106**, identify the tracking device **106** (e.g., using a unique identifier associated with the tracking device **106**), and/or detect a location associated with the tracking device **106** (e.g., a location of the first mobile community device **104***a* at the time of the communication with the tracking device **106**). This information can be used to provide updated locations and/or respond to a location request from the tracking system **100** regarding the tracking device **106**. In some embodiments, the steps performed by the first community mobile device **104***a* can be hidden from the first community user **105***a*. Accordingly, the first community mobile device **104***a* can assist in locating the tracking device **106** without bother and without the knowledge of the first community user **105***a*.

[0029] As mentioned above, the tracking system **100** can assist a user **103** in locating a tracking device **106**. The tracking device may be a chip, tile, tag, or other device for housing circuitry and that may be attached to or enclosed within an object such as a wallet, keys, purse, car, or other object that the user **103** may track. Additionally, the tracking device **106** may include a speaker for emitting a sound and/or a transmitter for broadcasting a beacon. In one configuration, the tracking device **106** may periodically transmit a beacon signal that may be detected using a nearby mobile device **102** and/or community mobile device **104**. In some configurations, the tracking device **106** broadcasts a beacon at regular intervals (e.g., one second intervals) that may be detected from a nearby mobile device (e.g., community mobile device **104**). The strength of the signal emitted from the tracking device **106** may be used to determine a degree of proximity to the mobile device **102** or community mobile device **104** that detects the signal. For example, a higher strength signal would indicate a close proximity between the tracking device **106** and the mobile device **102** and a lower strength signal would indicate a more remote proximity between the tracking device **106** and the mobile device **102**, though in some embodiments, the tracking device **106** can intentionally vary the transmission strength of the beacon signal. In some cases, the strength of signal or absence of a signal may be used to indicate that a tracking device **106** is lost.

System Overview

[0030] FIG. **2** illustrates an example tracking system for use in a tracking system environment, according to one embodiment. As shown, the tracking system **100** may include, but is not limited to, an association manager **204**, a tracking device location manager **206**, and a data manager **208**, each of which may be in communication with one another using any suitable communication technologies. It will be recognized that although managers **204-208** are shown to be separate in FIG. **2**, any of the managers **204-208** may be combined into fewer managers, such as into a single manager, or divided into more managers as may serve a particular embodiment.

[0031] The association manager **204** may be configured to receive, transmit, obtain, and/or update information about a user **103** and/or information about one or more specific tracking devices (e.g., tracking device **106**). In some configurations, the association manager **204** may associate information associated with a user **103** with information associated with a tracking device **106**. For example, user information and tracking information may be obtained by way of a mobile device **102**, and the association manager **204** may be used to link the user information and tracking information. The association between user **103** and tracking device **106** may be used for authentication purposes, or for storing user information, tracking device information, permissions, or other information about a user **103** and/or tracking device **106** in a database.

[0032] The tracking system **100** also includes a tracking device location manager **206**. The tracking device location manager **206** may receive and process an indication that the tracking device **106** is lost from a mobile device (e.g., mobile device **102** or community mobile devices **104**). For example, the tracking system **100** may receive a lost indication from a mobile device **102** indicating that the tracking device **106** is lost. The tracking device location manager **206** may set a flag on a database (e.g., tracker database **212**) indicating that the tracking device **106** is lost. The tracking device location manager **206** may also query a database to determine tracking information corresponding to the associated user **103** and/or tracking device **106**. The tracking system **100** may obtain tracking device information and provide the tracking device information or other information associated with the tracking device **106** to a plurality of community mobile devices **104** to be on alert for the lost or unavailable tracking device **106**.

[0033] The tracking device location manager **206** may also receive a location from one or more community mobile devices **104** that detect the tracking device **106**, for instance in response to the community mobile device receiving a beacon signal transmitted by the tracking device **106**, without the tracking device **106** having been previously marked as lost. In such embodiments, a user corresponding to the mobile device **102** can request a most recent location associated with the tracking device from the tracking system **100**, and the location manager **206** can provide the location received from the community mobile device for display by the mobile device **102**. In some embodiments, the location manager **206** provides the location of the tracking device **106** received from a community mobile device either automatically (for instance if the tracking device **106** is marked as lost) or at the request of a user of the mobile device **102** (for instance, via an application on the mobile device **102**). The location manager **206** can provide a location of a tracking device **106** to a mobile device **102** via a text message, push notification, application notification, automated voice message, or any other suitable form of communication. In some embodiments, the tracking device location manager **206** determines and stores a last known location for a tracking device **106** in response to the tracking device **106** disconnecting from a mobile device **102** or, in some implementations, a community mobile device **104**. The last known location for a tracking device **106** can be used as the most recent location associated with the tracking device **106** if the tracking device has not recently connected to a mobile device **102** or community mobile device **104**.

[0034] The tracking device location manager **206** may further manage providing indications about whether a tracking device **106** is lost or not lost. For example, as discussed above, the tracking device location manager **206** may provide a location request to the community of mobile devices **104** indicating that a tracking device **106** is lost. Additionally, upon location of the tracking device **106** by the user **103** or by one of the community of users **105**, the tracking device location manager **206** may provide an indication to the user **103**, community user **105**, or tracking system **100** that the tracking device **106** has been found, thus removing any flags associated with a tracking device and/or canceling any location request previously provided to the community of users **105**. For example, where a user **103** sends an indication that the tracking device **106** is lost to the tracking system **100** and later finds the tracking device **106**, the mobile device **102** may provide an indication to the tracking system **100** that the tracking device **106** has been found. In response, the tracking device location manager **206** may remove a flag indicating that the tracking device **106** is lost and/or provide an updated indication to the community of users **105** that the tracking device **106** has been found, thus canceling any instructions associated with the previously provided location request. In some configurations, the notification that the tracking device **106** has been found may be provided automatically upon the mobile device **102** detecting the tracking device **106** within a proximity of the mobile device **102**. Alternatively, the notification that the tracking device **106** has been found may be provided by the user **103** via user input on the mobile device **102**. In another example, a known user (e.g., a friend or family member) with whom the tracking device **106** has been shared may provide an indication that the tracking device **106** has been found.

[0035] The tracking system **100** additionally includes a data manager **208**. The data manager **208** may store and manage information associated with users, mobile devices, tracking devices, permissions, location requests, and other data that may be stored and/or maintained in a database related to performing location services of tracking devices. As shown, the data manager **208** may include, but is not limited to, a user database **210**, a tracker database **212**, permissions data **214**, and location request data **216**. It will be recognized that although databases and data within the data manager **208** are shown to be separate in FIG. **2**, any of the user database **210**, tracker database **212**, permissions data **214**, and location request data **216** may be combined in a single database or manager, or divided into more databases or managers as may serve a particular embodiment.

[0036] The data manager **208** may include the user database **210**. The user database **210** may be used to store data related to various users. For example, the user database **210** may include data about the user **103** as well as data about each user **105** in a community of users. The community of users **105** may include any user that has provided user information to the tracking system **100** via a mobile device **102**, **104** or other electronic device. The user information may be associated with one or more respective tracking devices **106**, or may be stored without an association to a particular tracking device. For example, a community user **105** may provide user information and permit performance of tracking functions on the community mobile device **104** without owning or being associated with a tracking device. The user database **210** may also include information about one or more mobile devices or other electronic devices associated with a particular user.

[0037] The data manager **208** may also include a tracker database **212**. The tracker database **212** may be used to store data related to tracking devices. For example, the tracker database **212** may include tracking data for any tracking device **106** that has been registered with the tracking system **100**. Tracking data may include unique tracker identifications (IDs) associated with individual tracking devices **106**. Tracker IDs may be associated with a respective user **103**. Tracker IDs may also be associated with multiple users. Additionally, the tracker database **212** may include any flags or other indications associated with whether a specific tracking device **106** has been indicated as lost and whether any incoming communications with regard to that tracking device **106** should be processed based on the presence of a flag associated with the tracking device **106**. Similarly, tracking data may include a last known location of the tracking device **106**.

[0038] The data manager **208** may further include permissions data **214** and location request data **216**. Permissions data **214** may include levels of permissions associated with a particular user **103** and/or tracking device **106**. For example, permissions data **214** may include additional users that have been indicated as sharing a tracking device **106**, or who have been given permission to locate or receive a location of a tracking device **106**. Location request data **216** may include information related to a location request or a lost indication received from the user **103** via a mobile device **102**.

[0039] FIG. **3** illustrates an example user mobile device for use in a tracking system environment, according to one embodiment. As shown, the mobile device **102** may include, but is not limited to, a user interface manager **302**, a location request manager **304**, a database manager **306**, and a tracking manager **308**, each of which may be in communication with one another using any suitable communication technologies. It will be recognized that although managers **302-308** are shown to be separate in FIG. **3**, any of the managers **302-308** may be combined into fewer managers, such as into a single manager, or divided into more managers as may serve a particular embodiment.

[0040] As will be explained in more detail below, the mobile device **102** includes the user interface manager **302**. The user interface manager **302** may facilitate providing the user **103** access to data on a tracking system **100** and/or providing data to the tracking system **100**. Further, the user interface manager **302** provides a user interface by which the user **103** may communicate with tracking system **100** and/or tracking device **106** via mobile device **102**.

[0041] The mobile device **102** may also include a location request manager **304**. The location request manager **304** may receive and process a request input to the mobile device **102** to send an indication that a tracking device **106** is lost to a tracking system **100**. For example, the user **103** may provide an indication that a tracking device **106** is lost, unreachable, or otherwise unavailable, from the mobile device **102** via the user interface manager **302**, and the location request manager

304 may process the lost indication and provide any necessary data to the tracking system **100** for processing and relaying a location request to other users **105** over a network **108**. In some configurations, an indication that a tracking device **106** is lost is provided via user input. Alternatively, the indication may be transmitted automatically in response to the mobile device **102** determining that a tracking device **106** is lost.

[0042] In addition, the location request manager **304** can request a location of the tracking device **106** without the tracking device **106** being identified as lost. For instance, a user can access a tracking device location feature of an application running on the mobile device **102** (for example, via the user interface manager **302**), and the location request manager **304** can request a most recent location of the tracking device **106** from the tracking system **100**. The location request manager **304** can receive the most recent location from the tracking system **100**, and can display the most recent location via the user interface manager **302**.

[0043] The mobile device **102** may also include a database manager **306**. The database manager **306** may maintain data related to the user **103**, tracking device **106**, permissions, or other data that may be used for locating a tracking device **106** and/or providing a request to a tracking system **100** for locating one or more tracking devices **106** associated with the user **103**. Further, the database manager **306** may maintain any information that may be accessed using any other manager on the mobile device **102**.

[0044] The mobile device **102** may further include a tracking manager **308**. The tracking manager **308** may include a tracking application (e.g., a software application) for communicating with and locating a tracking device **106** associated with the user **103**. For example, the tracking manager **308** may be one configuration of a tracking application installed on the mobile device **102** that provides the functionality for locating a tracking device **106** and/or requesting location of a tracking device **106** using a tracking system **100** and/or a plurality of community mobile devices **104**. As shown, the tracking manager **308** may include, but is not limited to, a Bluetooth Low Energy (BLE) manager **310**, a persistence manager **312**, a local files manager **314**, a motion manager **316**, a secure storage manager **318**, a settings manager **320**, a location manager **322**, a network manager **324**, a notification manager **326**, a sound manager **328**, a friends manager **330**, a photo manager **332**, an authentication manager **334**, and a device manager **336**. Thus, the tracking manager **308** may perform any of the functions associated with managers **310-338**, described in additional detail below.

[0045] The BLE manager **310** may be used to manage communication with one or more tracking devices **106**. The persistence manager **312** may be used to store logical schema information that is relevant to the tracking manager **308**. The local files manager **314** may be responsible for managing all files that are input or output from the mobile device **102**. The motion manager **316** may be responsible for all motion management required by the tracking manager **308**. The secure storage manager may be responsible for storage of secure data, including information such as passwords and private data that would be accessed through this sub-system. The settings manager **320** may be responsible for managing settings used by the tracking manager **308**. Such settings may be user controlled (e.g., user settings) or defined by the tracking manager **308** for internal use (e.g., application settings) by a mobile device **102** and/or the tracking system **100**. The location manager **322** may be responsible for all location tracking done by the tracking manager **308**. For example, the location manager **322** may manage access to the location services of the mobile device **102** and works in conjunction with other managers to persist data. The network manager **324** may be responsible for all Internet communications from the tracking manager **308**. For example, the network manager **324** may mediate all Internet API calls for the tracking manager **308**. The notification manager **326** may be responsible for managing local and push notifications required by the tracking manager **308**. The sound manager **328** may be responsible for playback of audio cues by the tracking manager **308**. The friends manager **330** may be responsible for managing access to contacts and the user's social graph. The photo manager **332** may be responsible for capturing and managing photos used by the tracking manager **308**. The authentication manager **334** may be responsible for handling the authentication (e.g., sign in or login) of users. The authentication manager **334** may also include registration (e.g., sign up) functionality. The authentication manager **334** further coordinates with other managers to achieve registration functionality. The device manager **336** may be responsible for managing the devices discovered by the tracking manager **308**. The device manager **336** may further store and/or maintain the logic for algorithms related to device discovery and update.

[0046] FIG. **4** illustrates an example community mobile device for use in a tracking system environment, according to one embodiment. As shown, the community mobile device **104** may include, but is not limited to, a user interface manager **402**, a tracking device manager **404**, a database manager **406**, and a tracking manager **408**, each of which may be in communication with one another using any suitable communication technologies. The user interface manager **402**, database manager **406**, and tracking manager **408** illustrated in FIG. **4** may include similar features and functionality as the user interface manager **302**, database manager **306**, and tracking manager **308** described above in connection with FIG. **3**. It will be recognized that although managers **402-408** are shown to be separate in FIG. **4**, any of the managers **402-408** may be combined into fewer managers, such as into a single manager, or divided into more managers as may serve a particular embodiment.

[0047] The community mobile device **104** may include a tracking device manager **404**. The tracking device manager **404** may facilitate scanning for nearby tracking devices **106**. In some configurations, the tracking device manager **404** can continuously or periodically scan (e.g., once per second) for nearby tracking devices **106**. The tracking device manager **404** may determine whether to provide an updated location of the nearby tracking device **106** to the tracking system **100**. In some configurations, the tracking device manager **404** provides a location of a nearby tracking device **106** automatically. Alternatively, the tracking device manager **404** may determine whether the location of the tracking device **106** has been recently updated, and may determine whether to provide an updated location based on the last time a location of the tracking device **106** has been updated (e.g., by the community mobile device **104**). For example, where the community mobile device **104** has provided a recent update of the location of a tracking device **106**, the tracking device manager **404** may decide to wait a predetermined period of time (e.g., 5 minutes) before providing an updated location of the same tracking device **106**.

[0048] In one configuration, the tracking device manager **404** may receive and process a location request or other information relayed to the community mobile device **104** by the tracking system **100**. For example, the tracking device manager **404** may receive an indication of a tracking device **106** that has been indicated as lost, and provide a location of the tracking device **106** if it comes within proximity of the community mobile device **104**. In some configurations, the community mobile device **104** is constantly scanning nearby areas to determine if there is a tracking device **106** within a proximity of the community mobile device **104**. Therefore, where a tracking device **106** that matches information provided by the tracking system **100** (e.g., from the location request) comes within proximity of the community mobile device **104**, the tracking device manager **404** may generate and transmit a response to the location request to the tracking system **100**, which may be provided to the user **103** associated with the tracking device **106**. Further, generating and transmitting the response to the tracking request may be conditioned on the status of the tracking device **106** being flagged as lost by the mobile device **102** and/or the tracking system **100**.

[0049] The tracking device manager **404** may additionally provide other information to the tracking system **100** in response to receiving the tracking request. For example, in addition to providing a location of the community mobile device **104**, the tracking device manager may provide a signal strength associated with the location to indicate a level of proximity to the location of the community mobile device **104** provided to the user **103**. For example, if a signal strength is high, the location provided to the user **103** is likely to be more accurate than a location accompanied by a low signal strength. This may provide additional information that the user **103** may find useful in determining the precise location of tracking device **106**.

[0050] As described above, the tracking device manager **404** may determine whether to send a location within the proximity of the tracking device **106** to the tracking system **100**. The determination of whether to send a location to the tracking system **100** may be based on a variety of factors. For example, a tracking device manager **404** may determine to send a location of the tracking device **106** to a tracking system **100** based on whether the detected tracking device **106** has been indicated as lost or if a tracking request has been provided to the community mobile device **104** for the particular tracking device **106**. In some configurations, the community mobile device **104** may send an update of a location of a tracking device **106** even if the tracking device **106** is not associated with a current tracking request or if the tracking device **106** is not indicated as lost. For example, where the location of a tracking device **106** has not been updated for a predetermined period of time, the community mobile device **104** may provide an update of a tracking device location to the tracking system **100**, regardless of whether a tracking request has been received.

[0051] In some configurations, the community mobile device **104** may include additional features. For example, the community mobile device **104** may allow a tracking system **100** to snap and download a photo using photo functionality of the community mobile device **104**. In some configurations, this may be an opt-in feature by which a community user **105** permits a tracking system **100** to take a snap-shot and possibly provide a visual image of an area within a proximity of the tracking device **106**.

[0052] FIG. **5** illustrates an example tracking device for use in a tracking system environment, according to one embodiment. The tracking device **106** of FIG. **5** includes an interface **502**, a transceiver **504**, a controller **506**, one or more sensors **508**, and a GPS unit **510**. The transceiver **504** is a hardware circuit capable of both transmitting and receiving signals. It should be noted that in other embodiments, the tracking device **106** includes fewer, additional, or different components than those illustrated in FIG. **5**. For instance, tracking devices might not include the GPS unit **510** and can still implement the functionalities described herein.

[0053] The interface **502** provides a communicative interface between the tracking device **106** and one or more other devices, such as a mobile device **102**. For instance, the interface **502** can instruct the transceiver **504** to output beacon signals as described above (for example, periodically or in response to a triggering event, such as a detected movement of the tracking device **106**). The interface **502** can, in response to the receiving of signals by the transceiver **504** from, for instance, the mobile device **102**, manage a pairing protocol to establish a communicative connection between the tracking device **106** and the mobile device **102**. As noted above, the pairing protocol can be a BLE connection, though in other embodiments, the interface **502** can manage other suitable wireless connection protocols (such as WiFi, Global System for Mobile Communications or GSM, and the like).

[0054] The controller **506** is a hardware chip that configures the tracking device **106** to perform one or more functions or to operate in one or operating modes or states. For instance, the controller **506** can configure the interval at which the transceiver broadcasts beacon signals, can authorize or prevent particular devices from pairing with the tracking device **106** based on information received from the devices and permissions stored at the tracking device, can increase or decrease the transmission strength of signals broadcasted by the transceiver, can configure the interface to emit a ringtone or flash an LED light, can enable or disable various tracking device sensors, can enable or disable a tracking device GPS unit, can enable or disable communicative functionality of the tracking device **106** (such as a GSM transmitter and receiving), can configure the tracking device into a sleep mode or awake mode, can configure the tracking device into a power saving mode, and the like. The controller **506** can configure the tracking device to perform functions or to operate in a particular operating mode based on information or signals received from a device paired with or attempting to pair with the tracking device **106**, based on an operating state or connection state of the tracking device **106**, based on user-selected settings, based on information stored at the tracking device **106**, based on a detected location of the tracking device **106**, based on historical behavior of the tracking device **106** (such as a previous length of time the tracking device was configured to operate in a particular mode), based on information received from the sensors **508** or the GPS unit **510**, or based on any other suitable criteria.

[0055] The sensors **508** can include motion sensors (such as gyroscopes or accelerators), altimeters, orientation sensors, proximity sensors, light sensors, or any other suitable sensor configured to detect an environment of the tracking device **106**, a state of the tracking device **106**, a movement or location of the tracking device **106**, and the like. The sensors **508** are configured to provide information detected by the sensors to the controller **506**. The GPS unit **510** is configured to detect a location of the tracking device **106** based on received GPS signals, and is configured to provide detected locations to the controller **506**.

Last Known Location Determination

[0056] As described herein, determining and storing a last known location for a tracking device **106** can allow the tracking system **100** to provide an approximate location of a tracking device **106** to one or more mobile devices **102** or community mobile devices **104**, even if the tracking device **106** has not recently connected to a mobile device **102** or community mobile device **104**. In some embodiments, the mobile device **102** and community mobile device **104** report events including location data to the tracking system **100**. For example, mobile devices can report events to the tracking system **100** periodically, or in response to any suitable trigger, such as connecting to a tracking device **106**, disconnecting from a tracking device **106**, if location data becomes available for the mobile device **102**, or for any other suitable reason. For example, a mobile device **102** (or community mobile device **104**) can report a disconnection event to the tracking system **100** when the mobile device **102** disconnects from a tracking device **106**. A disconnection event can include various information about the mobile device **102** and the tracking device **106** at the time the tracking device **106** disconnected from the mobile device **102**. For example, a disconnection event can contain, for example, a timestamp, tracking device ID, location of the mobile device **102** when the disconnection event was reported, and a location accuracy of the location. A location may be reported in any suitable format, such as a set of latitude-longitude coordinates.

[0057] In some implementations, the tracking system **100** uses the location of the mobile device **102** included in the disconnection event as the last known location of the tracking device **106**. In some implementations, because the mobile device **102** was connected to the tracking device **106**, the location of the mobile device **102** can be used as an estimate of the location of the tracking device **106** at that time. In some cases, the location reported with the disconnection event accurately represents the last known location of the tracking device **106**. However, the location reported in the disconnection event is not always the most accurate available location for the tracking device **106**, nor do all disconnection events contain a useable location. For example, location services may not be available for the mobile device **102** when the tracking device **106** disconnects, or location data for the mobile device **102** can be inconsistent or inaccurate when the tracking device **106** disconnects (for example, if the mobile device **102** is in a parking garage which interferes with GPS signals). Therefore, in some embodiments, determining the last known location for a tracking device **106** can be based on reported location information from a variety of sources or otherwise require more calculation than finding the most recently reported location for the tracking device **106**.

[0058] In some embodiments, locations reported by the mobile device **102** before or after the disconnect event can be used to infer a last known location of the tracking device **106** at the time of the disconnection event. Similarly, a last known location for a tracking device **106** can be determined or improved based on locations reported by other mobile devices **102** or community mobile devices **104** which recently connected to the tracking device **106**.

[0059] FIG. **6** illustrates an example environment for determining a tracking device location based on later mobile device location updates, according to one embodiment. The environment **600** of FIG. **6** includes an initial device location **610** with a corresponding initial location accuracy **615**, a later device location **620** with a corresponding with a later location accuracy **625**, and an estimate location accuracy **630**.

[0060] In some embodiments, the initial device location **610** represents a location of the mobile device **102** at an initial time, for example, a location reported by the mobile device **102** in a disconnection event. The initial device location **610** is associated with a corresponding initial location accuracy **615** representing the potential deviation of the actual position of the mobile device **102** from the reported initial device location **610**. The initial location accuracy **615** can be any suitable measure of the accuracy of the reported initial device location **610**, for example an expected deviation from the initial device location **610** or a radius of a circle within which the mobile device **102** is expected to be located. The accuracy of a reported mobile device location can depend on available location services (for example, if a mobile device **102** has GPS disabled, but a rough location is determined based on nearby WiFi networks), natural variations or fluctuations in the reported location, environmental factors (for example, if the mobile device **102** is located in a parking structure or other building interfering with GPS signals), or any other suitable factor.

[0061] Over time, the mobile device **102** can report additional locations to the tracking system **100** in addition to the initial mobile device location **610**, according to some embodiments. For example, if the initial device location **610** was reported in a disconnection event, the mobile device **102** can continue to report location information to the tracking system **100** in other events. A mobile device **102** can report a location to the tracking system **100** in response to a second disconnection event (associated with a different tracking device **106**), periodically, while connected to a different tracking device **106**, when connecting to the tracking system **100**, or for any other suitable reason. In some implementations, each location reported by a mobile device **102** is associated with a location accuracy, which can vary based on, for example, environmental factors or natural variations as described above. The later device location **620** and later location accuracy **625** represent a relatively high accuracy location of the mobile device **102** reported at a time after the initial device location **610** was reported by the mobile device **102**. For example, the later device location **620** can represent the location of the mobile device **102** 25 seconds after the initial device location **610**. In the embodiment of FIG. **6**, the later device location **620** is more accurate than the initial mobile device location **610**. For example, this could be due to the mobile device **102** reacquiring a GPS signal between the time the initial device location **610** was reported to the tracking system **100** and the time the later device location **620** was reported to the tracking system **100**.

[0062] When a later device location **620** is more accurate than the initial device location **610**, the later device location **620** can be used to generate a more accurate estimate of the location of the mobile device **102** at the time the initial device location **610** was reported. Later device location **620** can be used as the estimated location of the mobile device **102** at the initial time. In some embodiments, to determine the estimated location accuracy **630**, the later location accuracy **625** is adjusted to account for the difference in time between the initial device location **610** and the later device location **620**. For example, the estimated location accuracy **630** can take into account potential motion of the mobile device **102** in the intervening time between the initial device location **610** and the later device location **620**. The later location accuracy **625** can be adjusted based on any suitable factor to generate the estimated location accuracy **630**, for example the difference in time between the initial and later device locations **610** and **620**, the difference in location between the initial and later device locations **610** and **620**, or any other suitable factor.

[0063] In the embodiment of FIG. **6**, the estimated location accuracy **630** represents the adjusted accuracy of the estimated device location. In some embodiments, the estimated location accuracy **630** is more accurate than the initial location accuracy **615** (even after the adjustment). Therefore, the later device location **620** and estimated location accuracy **630** can be used as the last known location of the tracking device **106**. Similar techniques can be used to determine a last known location for a tracking device **106** from location information preceding the disconnection event. In some embodiments, to be used as the last known location of a tracking device **106**, the later location accuracy **625** of the later device location **620** should be greater than a threshold accuracy and the later device location **620** should be reported within a threshold time of the disconnection event.

[0064] FIG. **7** illustrates an example environment for determining a tracking device location based on multiple reported locations, according to one embodiment. The environment **700** of FIG. **7** includes a mobile device **710** and associated mobile device location accuracy **715**, a community mobile device A **720** and associated community mobile device A location accuracy **725**, a community mobile device B **730** and associated community mobile device B location accuracy **735**, and an estimated tracking device location **740** determined based on the reported locations.

[0065] In the embodiment of FIG. **7**, the mobile device **710** and the community mobile devices A **720** and B **730** each report (to the tracking system **100**) a location associated with the same tracking device **106**. Each reported location can be associated with other relevant information, for example, a timestamp, a signal strength of the tracking device **106** at the time the location was reported, and/or an accuracy of the reported location (for example, the mobile device location accuracy **715** and the community mobile device location accuracies A **725** and B **735**). In some embodiments, reported locations used to determine the location of a tracking device **106** can be associated with various times, within a certain threshold time period of the time at which to estimate the location.

[0066] To estimate a location for a tracking device **106** from the known locations of mobile devices **102** and community mobile devices **104** connected to the tracking device **106**, a location determination algorithm can be used, according to some embodiments. Each reported location (for example, the locations of the mobile device **710** and the community mobile devices A **720** and B **730**) can be assigned a weight in the location determination algorithm based on any suitable factors, for example the timestamp of the reported location, the accuracy of the reported location, and/or the strength of signal of the tracking device **106** at the mobile device **102** (or community mobile device **104**) when the location was reported. Any suitable location determination algorithm can be used to determine the estimated tracking device location **740** based on one or more reported locations. For example, a triangulation algorithm can be used to determine the estimated tracking device location **740**, such as a Delaunay triangulation operation, a greedy triangulation operation, a jump-and-walk triangulation operation, a kinetic triangulation operation, or the like. In other embodiments, the estimated tracking device location **740** can be determined by a weighted average of the reported locations of the mobile device **710** and the community mobile devices A **720** and B **730**, where each reported location is weighted based on one or more of the location accuracy, signal strength, and time of the reported location. In some embodiments, the accuracy of the estimated location can be calculated by similar methods.

[0067] In order to determine a last known location for a tracking device **106**, additional data reported about the tracking device **106** can be collected and referenced to determine the last known location. However, a tracking system **100** may deal with large numbers of tracking devices **106**, each associated with multiple mobile devices **102** and community mobile devices **104** which in turn report events (such as disconnection events) including location information and other relevant information the tracking system **100** in a continuous stream of data. Collecting the relevant data needed to determine a last known location of a tracking device based on each disconnection event can require the tracking system to filter and organize a large volume of data. Therefore, a “streaming” data system can be utilized to receive and organize relevant portions of data received at the tracking system **100** from a plurality of mobile devices **102** and community mobile device **104** to determine a last known location of a tracking device **106** for each disconnection event. In some implementations, a data streaming framework (for example, Apache Spark streaming) can be used to filter and organize incoming data (including disconnection requests and location updates) to determine a last known location. In some implementations, the data streaming framework allows the tracking system **100** to batch incoming data based on time and process each batch of data individually. For example, the incoming data stream can be separated and processed in batches of, for example, 20 seconds, or any other reasonable duration. In some embodiments, the duration of each batch is configurable, for example from 15-25 seconds.

[0068] FIG. **8** is a flowchart illustrating an example process for filtering data for determining a last known location of a tracking device, according to one embodiment. In some implementations, the process **800** is followed for each batch of data reported as part of a continuous data stream. The process **800** begins when a stream of tracking device event information is received **810** at the tracking system. Then, the tracking system can separate **820** events containing location updates from the received stream. For example, location updates can comprise the location of a tracking device **106**, mobile device **102**, or community mobile device **104** and may be reported in the context of a disconnection event (or any other suitable event). In some implementations, disconnection events are also separately handled (for example at step **830**) and are not separated as part of step **820**. Information unnecessary for calculating a last known location of a tracking device **106** can then be removed **825** from the separated location update data. For example, each event containing a device location update can be reduced to a timestamp of the event, an identifier of the reporting mobile device **102** (or community mobile device **104**), an identifier of a connected tracking device **106** (if present), a location of the reporting mobile device **102**, a location accuracy of the reported location, a signal strength of the tracking device **106** at the reporting mobile device **102**, and any other relevant information. For example, a MapReduce operation can be used to reduce location updates to only relevant information for calculating a last known location. In some implementations, filtering each tracking device location update to remove unnecessary information improves the performance of the tracking system by reducing the volume of data needed to be stored to determine the last known location of a tracking device **106**.

[0069] Disconnection events can be similarly separated **830** from the received data stream. Each disconnection event is then associated and grouped **840** with relevant tracking device location updates. For example, the tracking system **100** can associate each tracking device disconnection event with the location updates associated with the same tracking device **106** as the disconnection event and/or any other location updates from the reporting mobile device **102**. The grouped location updates, including any location updates from the disconnection event itself, can then be used to determine **850** the last known location for the disconnected tracking device **106**.

[0070] FIG. **9** is a flowchart illustrating an example process for determining the last known location of a tracking device, according to one embodiment. The process **900** begins when a disconnection event and associated location is received **910**, for example at a tracking system. In some implementations, the disconnection event includes location data, but the associated location data can also comprise additional location data gathered from other mobile devices or from different points in time than the disconnection event. For example, FIG. **8** discloses a process for filtering and organizing disconnection events with relevant location information. If relevant location data is available **920**, the process **800** can proceed with determining the last known location. For example, associated location data may need to be within a certain time window of the disconnection event, or have a threshold level of accuracy to be used to generate a last known location. In some embodiments, if no suitable location data is available for the disconnection event, no last known location can be determined for the tracking device.

[0071] In the process **900**, if multiple mobile devices (or community mobile devices) are reporting location data **930** for the tracking device, the reported location from each reporting mobile device can be combined **940** to determine the last known location of the tracking device. For example, a triangulation algorithm or weighted average of the reported locations can be used, as described in relation to FIG. **7**. In some implementations, only one location from each reporting mobile device is used to calculate the last known location of the tracking device (for example, the highest

accuracy or the reported location closest in time to the disconnection event).
[0072] In other cases, only one mobile device (or community mobile device) report relevant location data **930** about the tracking device. In these cases, the location (or locations) reported by the mobile device can be used to calculate the last known location. First, the tracking system can modify **950** the accuracy of each reported location based on the time difference between the location and the disconnection event (and any other suitable factor). After each location is modified to account for the time different, the tracking system can select **955** the best and/or most accurate location as the last known location of the tracking device.
[0073] In either case, the last known location of the tracking device is returned **960**. For example, the last known location can then be stored within the tracking system for later use, sent to an owner of the tracking device, or used for any other suitable purpose.
Additional Considerations
[0074] The foregoing description of the embodiments of the invention has been presented for the purpose of illustration; it is not intended to be exhaustive or to limit the invention to the precise forms disclosed. Persons skilled in the relevant art can appreciate that many modifications and variations are possible in light of the above disclosure.
[0075] Any of the devices or systems described herein can be implemented by one or more computing devices. A computing device can include a processor, a memory, a storage device, an I/O interface, and a communication interface, which may be communicatively coupled by way of communication infrastructure. Additional or alternative components may be used in other embodiments. In particular embodiments, a processor includes hardware for executing computer program instructions by retrieving the instructions from an internal register, an internal cache, or other memory or storage device, and decoding and executing them. The memory can be used for storing data or instructions for execution by the processor. The memory can be any suitable storage mechanism, such as RAM, ROM, flash memory, solid state memory, and the like. The storage device can store data or computer instructions, and can include a hard disk drive, flash memory, an optical disc, or any other suitable storage device. The I/O interface allows a user to interact with the computing device, and can include a mouse, keypad, keyboard, touch screen interface, and the like. The communication interface can include hardware, software, or a combination of both, and can provide one or more interfaces for communication with other devices or entities.
[0076] Some portions of this description describe the embodiments of the invention in terms of algorithms and symbolic representations of operations on information. These algorithmic descriptions and representations are commonly used by those skilled in the data processing arts to convey the substance of their work effectively to others skilled in the art. These operations, while described functionally, computationally, or logically, are understood to be implemented by computer programs or equivalent electrical circuits, microcode, or the like. Furthermore, it has also proven convenient at times, to refer to these arrangements of operations as modules, without loss of generality. The described operations and their associated modules may be embodied in software, firmware, hardware, or any combinations thereof.
[0077] Any of the steps, operations, or processes described herein may be performed or implemented with one or more hardware or software modules, alone or in combination with other devices. In one embodiment, a software module is implemented with a computer program product comprising a computer-readable medium containing computer program code, which can be executed by a computer processor for performing any or all of the steps, operations, or processes described.
[0078] Embodiments of the invention may also relate to an apparatus for performing the operations herein. This apparatus may be specially constructed for the required purposes, and/or it may comprise a general-purpose computing device selectively activated or reconfigured by a computer program stored in the computer. Such a computer program may be stored in a non-transitory, tangible computer readable storage medium, or any type of media suitable for storing electronic

instructions, which may be coupled to a computer system bus. Furthermore, any computing systems referred to in the specification may include a single processor or may be architectures employing multiple processor designs for increased computing capability.
[0079] Embodiments of the invention may also relate to a product that is produced by a computing process described herein. Such a product may comprise information resulting from a computing process, where the information is stored on a non-transitory, tangible computer readable storage medium and may include any embodiment of a computer program product or other data combination described herein.
[0080] Finally, the language used in the specification has been principally selected for readability and instructional purposes, and it may not have been selected to delineate or circumscribe the inventive subject matter. It is therefore intended that the scope of the invention be limited not by this detailed description, but rather by any claims that issue on an application based hereon. Accordingly, the disclosure of the embodiments of the invention is intended to be illustrative, but not limiting, of the scope of the invention, which is set forth in the following claims.

Claims

1. A method, comprising: receiving, by a tracking system server, a stream of tracking device event information associated with a plurality of tracking devices; extracting, by the tracking system server, and from the stream of tracking device event information, tracking device event information including location updates; filtering, by the tracking system server, the tracking device event information by removing information unrelated to any calculation of a last known location of at least one tracking device of the plurality of tracking devices; associating, by the tracking system server, the filtered tracking device event information with one or more disconnection events associated with the at least one tracking device; and identifying, by the tracking system server, and based at least in part on the filtered tracking device event information and the one or more disconnection events, the last known location of the at least one tracking device.
2. The method of claim 1, wherein receiving the stream of tracking device event information comprises receiving, by the tracking system server, a continuous stream of disconnection events and additional location information associated with the plurality of tracking devices.
3. The method of claim 1, wherein the location updates comprise one or more of a location of the at least one tracking device, a location of at least one mobile computing device associated with the plurality of tracking devices, or a location of at least one community mobile computing device associated with the tracking system server.
4. The method of claim 1, wherein extracting the tracking device event information comprises extracting, by the tracking system server, and from the stream of tracking device event information, batches of tracking device event information each defined over a time interval of an N number of seconds.
5. The method of claim 4, further comprising processing, by the tracking system server, each of the batches of the tracking device event information individually.
6. The method of claim 1, wherein filtering the tracking device event information further comprises filtering, by the tracking system server, the tracking device event information to include only one or more of a timestamp of the one or more disconnection events, an identifier of a mobile computing device having reported the one or more disconnection events, an identifier of the at least one tracking device, a location of the mobile computing device having reported the one or more disconnection events, a location accuracy of a reported location, or a signal strength of the at least one tracking device at the mobile computing device having reported the one or more disconnection events.
7. The method of claim 1, wherein filtering the tracking device event information further comprises filtering, by the tracking system server, the tracking device event information utilizing a

MapReduce process.
8. A tracking system server, comprising: one or more non-transitory computer-readable storage media including instructions; and one or more processors coupled to the one or more non-transitory computer-readable storage media, the one or more processors configured to execute the instructions to: receive a stream of tracking device event information associated with a plurality of tracking devices; extract, from the stream of tracking device event information, tracking device event information including location updates; filter the tracking device event information by removing information unrelated to any calculation of a last known location of at least one tracking device of the plurality of tracking devices; associate the filtered tracking device event information with one or more disconnection events associated with the at least one tracking device; and identify, based at least in part on the filtered tracking device event information and the one or more disconnection events, the last known location of the at least one tracking device.
9. The tracking system server of claim 8, wherein the instructions to receive the stream of tracking device event information further comprise instructions to receive a continuous stream of disconnection events and additional location information associated with the plurality of tracking devices.
10. The tracking system server of claim 8, wherein the location updates comprise one or more of a location of at least one tracking device, a location of at least one mobile computing device associated with the plurality of tracking devices, or a location of at least one community mobile computing device associated with the tracking system server.
11. The tracking system server of claim 8, wherein the instructions to extract the tracking device event information further comprise instructions to extract, from the stream of tracking device event information, batches of the tracking device event information each defined over a time interval of an N number of seconds.
12. The tracking system server of claim 11, wherein the instructions further comprise instructions to process each of the batches of the tracking device event information individually.
13. The tracking system server of claim 8, wherein the instructions to filter the tracking device event information further comprise instructions to filter the tracking device event information to include only one or more of a timestamp of the one or more disconnection events, an identifier of a mobile computing device having reported the one or more disconnection events, an identifier of the at least one tracking device, a location of the mobile computing device having reported the one or more disconnection events, a location accuracy of a reported location, or a signal strength of the at least one tracking device at the mobile computing device having reported the one or more disconnection events.
14. The tracking system server of claim 8, wherein the instructions to filter the tracking device event information further comprise instructions to filter the tracking device event information utilizing a MapReduce process.
15. A non-transitory computer-readable medium comprising instructions that, when executed by one or more processors of a tracking system server, cause the tracking system server to: receive a stream of tracking device event information associated with a plurality of tracking devices; extract, from the stream of tracking device event information, tracking device event information including location updates; filter the tracking device event information by removing information unrelated to any calculation of a last known location of at least one tracking device of the plurality of tracking devices; associate the filtered tracking device event information with one or more disconnection events associated with the at least one tracking device; and identify, based at least in part on the filtered tracking device event information and the one or more disconnection events, the last known location of the at least one tracking device.
16. The non-transitory computer-readable medium of claim 15, wherein the instructions to receive the stream of tracking device event information further comprise instructions to receive a continuous stream of disconnection events and additional location information associated with the

plurality of tracking devices.

17. The non-transitory computer-readable medium of claim 15, wherein the location updates comprise one or more of a location of at least one tracking device, a location of at least one mobile computing device associated with the plurality of tracking devices, or a location of at least one community mobile computing device associated with the tracking system server.

18. The non-transitory computer-readable medium of claim 15, wherein the instructions to extract the tracking device event information further comprise instructions to extract, from the stream of tracking device event information, batches of the tracking device event information each defined over a time interval of an N number of seconds.

19. The non-transitory computer-readable medium of claim 18, wherein the instructions further comprise instructions to process each of the batches of the tracking device event information individually.

20. The non-transitory computer-readable medium of claim 15, wherein the instructions to filter the tracking device event information further comprise instructions to filter the tracking device event information to include only one or more of a timestamp of the one or more disconnection events, an identifier of a mobile computing device having reported the one or more disconnection events, an identifier of the at least one tracking device, a location of the mobile computing device having reported the one or more disconnection events, a location accuracy of a reported location, or a signal strength of the at least one tracking device at the mobile computing device having reported the one or more disconnection events.
